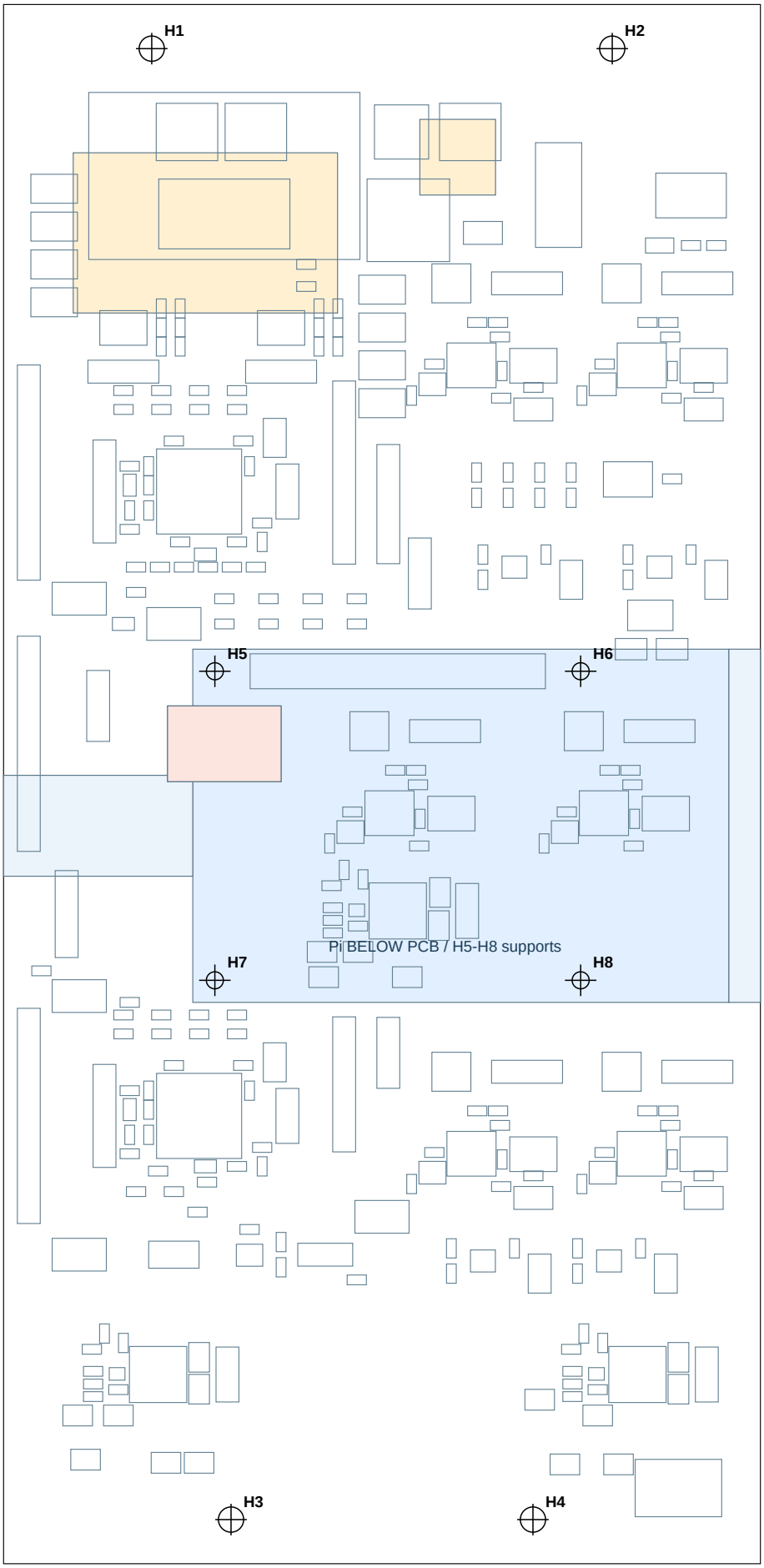
Lab 4 - revised PCB fit template

Print at 100% / Actual size. Check the 50 mm bar before comparing with the car.

SELECTED CAD HOLE CENTERS - PLACEMENT PREVIEW - NOT FOR FABRICATION

REAR / DRIVE MOTORS - new rear pair A1/A2



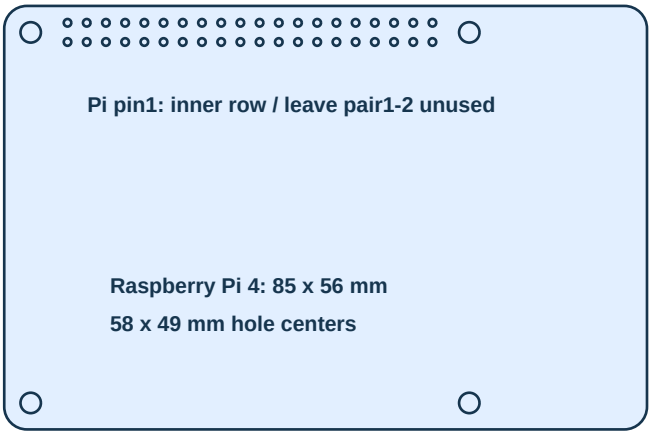
FRONT / STEERING WHEELS

This bar must measure 50 mm.

H1-H4: 4.0 mm PCB drills for M3. Rear chassis holes: 3.2 mm in CAD. PCB: 120 x 247.212 mm.

Pi alignment and vertical fit check

Top view below is 1:1. Place the unpowered Pi component-side-up to compare holes and pins.



USB / Ethernet face right.

GPIO row faces the REAR.

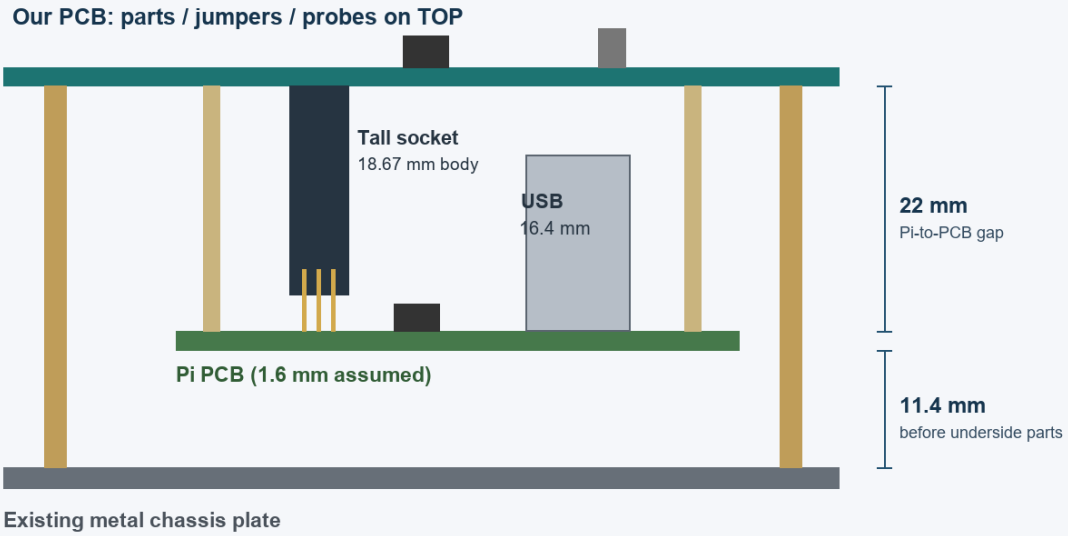
Socket mates Pi pins3-38 only.

Do not use this as a drill guide.

Main-PCB Pi origin: (30,89) mm.

How the boards stack - proposed dimensions

Pi component side faces UP. J70 socket is soldered to the UNDERSIDE of our PCB.



35 mm chassis standoffs proposed. Caliper heights recorded; final mating is checked at assembly.

Recorded geometry and remaining fit checks:

Chassis spacers: 35 mm nominal; four-hole paper fit passed with ~50 mm spacers.

Pi-to-PCB gap: 22 mm nominal; calculated socket engagement 5.17 mm.

Reported underside parts <5 mm; calculated nominal metal clearance 6.4 mm.

USB top measured ~18 mm from Pi board bottom; check any added cooler.

Measured chassis pattern retained per team-relayed Ayaan decision; 4 mm drills.

Still check unknown plug/cable/steering envelopes and assembled socket fit.

Pi hardware OD <=5.5 mm. Chassis / Pi edge exclusions: 4 / 3 mm. Socket first pair aligns to Pi3/4.